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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Merge Sort
- c. Heap Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It increases the number of comparisons needed
- b. It has no effect on the search operation
- c. It makes the search operation slower
- d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Sorting data in a non-sequential manner
c. Using the divide-and-conquer approach
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It divides the list into two halves, sorts each half, and then merges them
b. It compares adjacent elements and swaps them if necessary
c. It builds a sorted array one element at a time
d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Dividing the input into parts, solving each part, and combining the solutions
b. Rearranging data without sorting
c. Sorting data sequentially
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **7**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Selection
c. Searching
d. All of the above

The correct answer is: All of the above

Question **8**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Searching
c. Merging
d. Sorting

The correct answer is:
Complexity

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. arrange()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Merging two datasets
- c. Removing duplicates from a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It sorts data out of the box efficiently
- c. It is less efficient than custom sorting algorithms
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Makes data harder to manage
c. Decreases the efficiency of algorithms
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort